Run report

2026-10-27T143728Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	500,000
dim	1	dataset	gaussian
seed	1,426,827,473	data_seed	42
hidden_units	64	n_layers	2
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	5.202×10^{-3}	best_epoch	98
compile_cache_warm	True		
gaussian_params	dim=1, mu_gen=[0.5], mu_true=[0], cov_gen=[[0.81]], cov_true=[[1]], cov_detector=[[0.25]]		
mmd_sigmas_detector	$0.7470 \times (\frac{1}{2},\ \frac{1}{\sqrt{2}},\ 1,\ \sqrt{2},\ 2)$		
mmd_sigmas_particle	$0.6688 \times (\frac{1}{2},\ \frac{1}{\sqrt{2}},\ 1,\ \sqrt{2},\ 2)$		

Timing

Phase	Seconds	Share (%)	Detail
DECONVOLVE			
data	0.071	0.1	gaussian cache hit – train
train	18.884	19.2	train
transfer	6.451		train
compile	2.907		train
epochs	6.269		train
select	2.613		train
particle_mmd	0.448	0.5	train
save	0.447	0.5	train
plots	74.758	76.1	train
evaluate	3.573	3.6	train
total	98.180	100.0	
OMNIFOLD baseline			
parse_config	0.002	0.0	omnifold
data	0.074	0.0	gaussian cache hit – omnifold
omnifold	340.241	98.4	niter=3, epochs=50 – omnifold
init	3.667		DataLoaders and the two MLPs – omnifold
unfold	288.225		MultiFold.Unfold – omnifold
iter1_step1	66.154		detector-level reweighting – omnifold
iter2_step1	43.693		detector-level reweighting – omnifold
iter3_step1	43.570		detector-level reweighting – omnifold
iter1_step2	38.763		particle-level reweighting – omnifold
iter2_step2	37.478		particle-level reweighting – omnifold
iter3_step2	58.533		particle-level reweighting – omnifold
reweight	1.460		evaluating weights on z_target – omnifold
evaluate	5.295	1.5	1 variables – omnifold
total	345.612	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein Distance [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	507.8	—	—	**16.43**	**96.8**	95.23	81.2

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	28.91	—	—	**0.6968**	**97.6**	1.748	94.0

VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	111.5	—	—	**2.560**	**97.7**	6.707	94.0

Sliced Wasserstein, joint [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
All	455.0	—	—	**14.72**	**96.8**	85.32	81.2

Mean 1D distance over random directions, in Data standard deviations. IBU: product of its per-observable weights.

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	508.5	—	—	**15.65**	**96.9**	94.87	81.3

JS divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	37.03	—	—	**0.7575**	**98.0**	2.365	93.6

VLC divergence [$\times 10^3$]

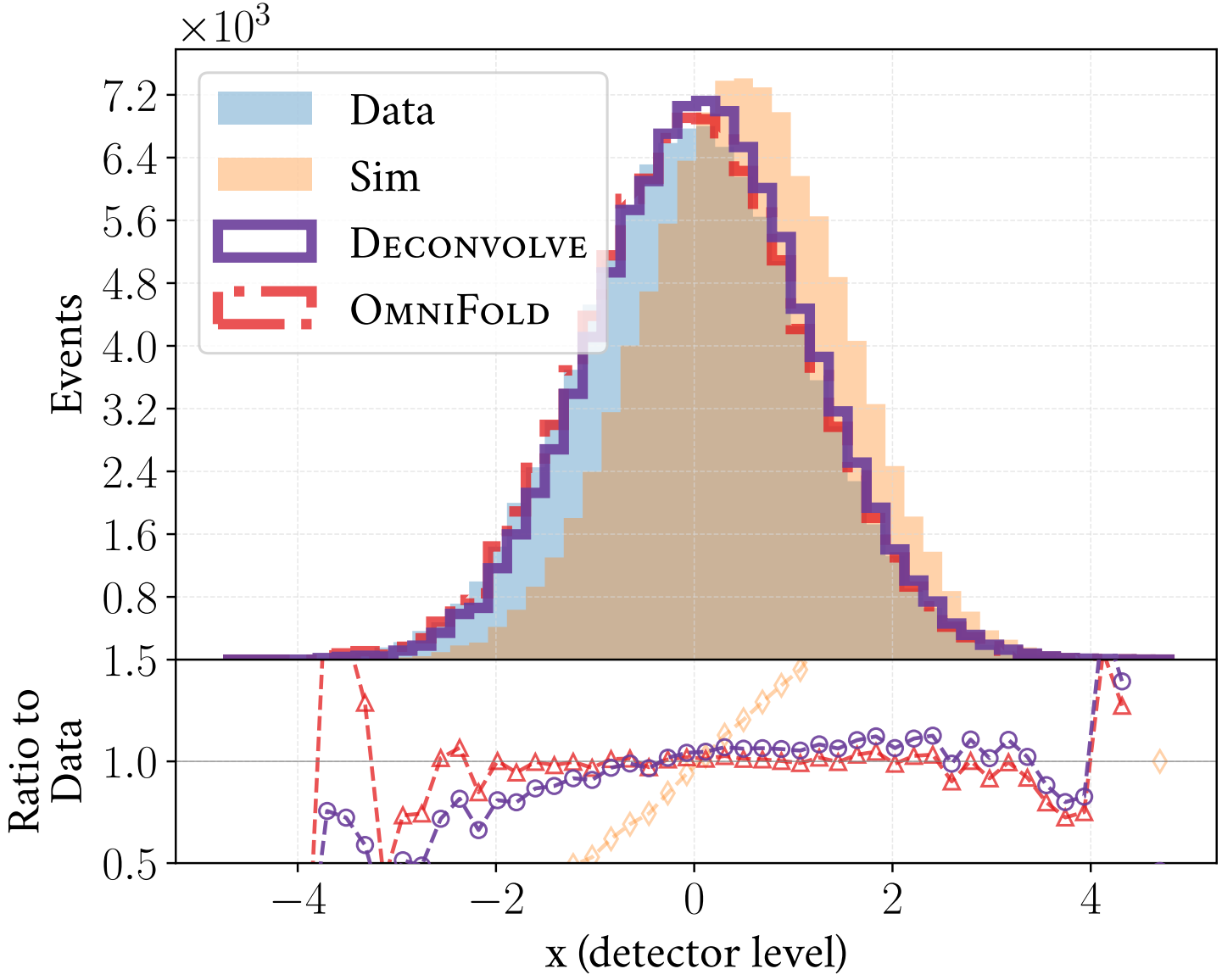
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
dim_0	141.3	—	—	**2.819**	**98.0**	9.092	93.6

Sliced Wasserstein, joint [$\times 10^3$]

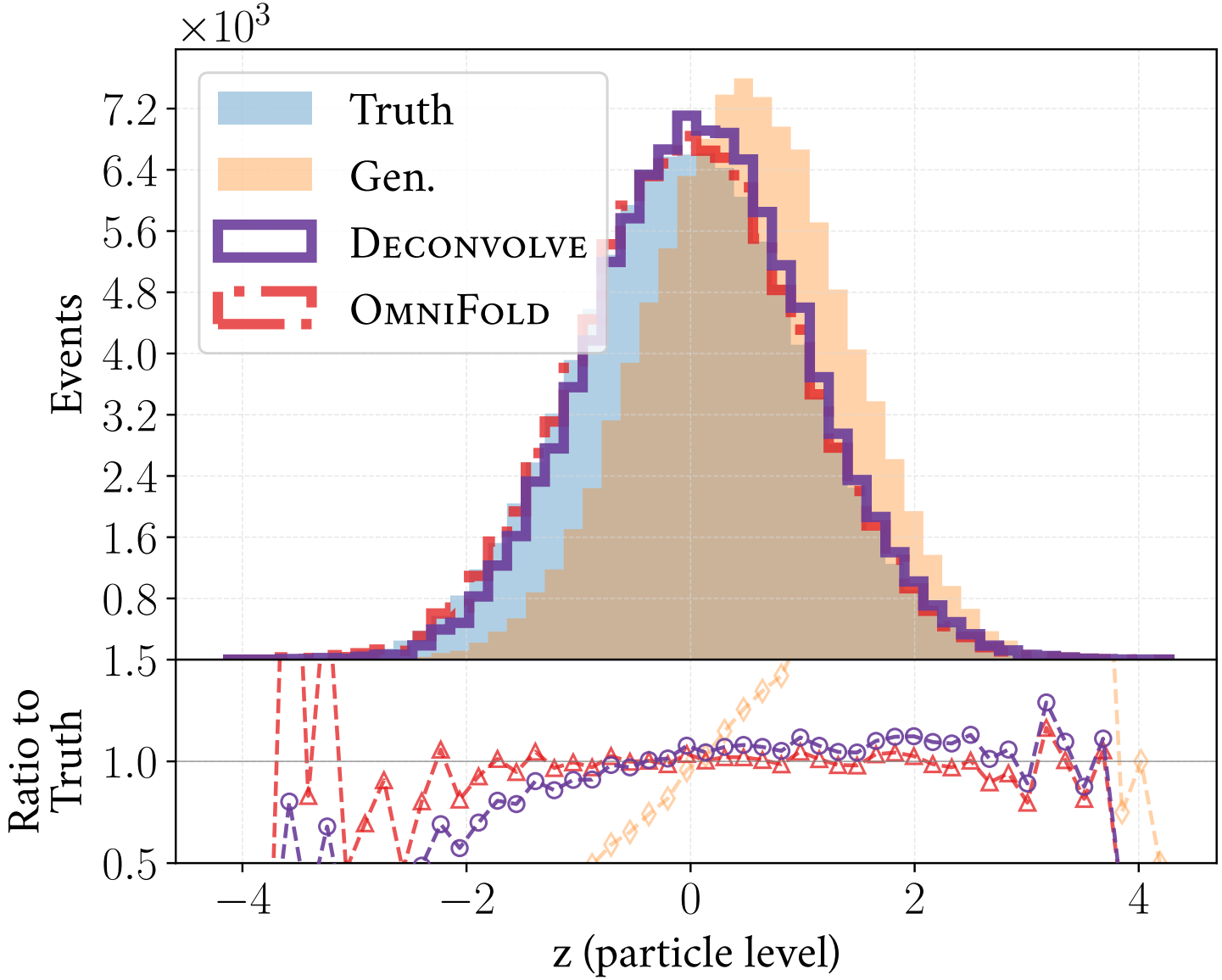
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	Deconvolve	impr. (%)
All	508.4	—	—	**15.65**	**96.9**	94.87	81.3

Mean 1D distance over random directions, in Truth standard deviations. IBU: product of its per-observable weights.

Detector Level



Particle Level



Training diagnostics

Training losses



Selection criteria

